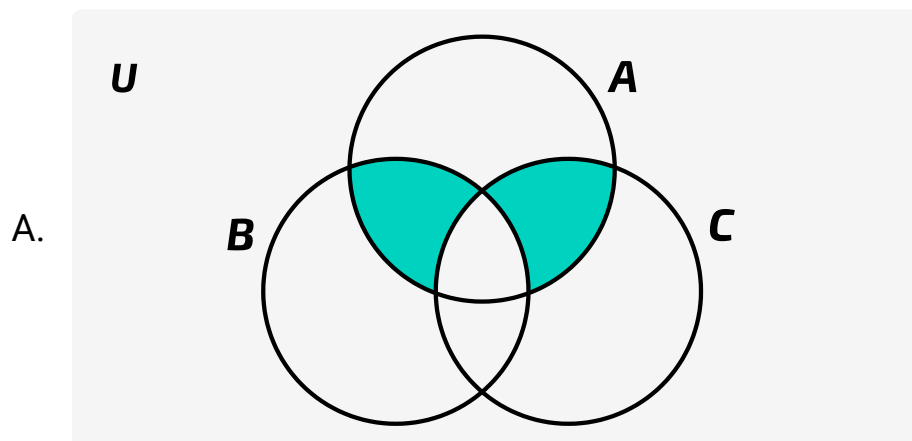


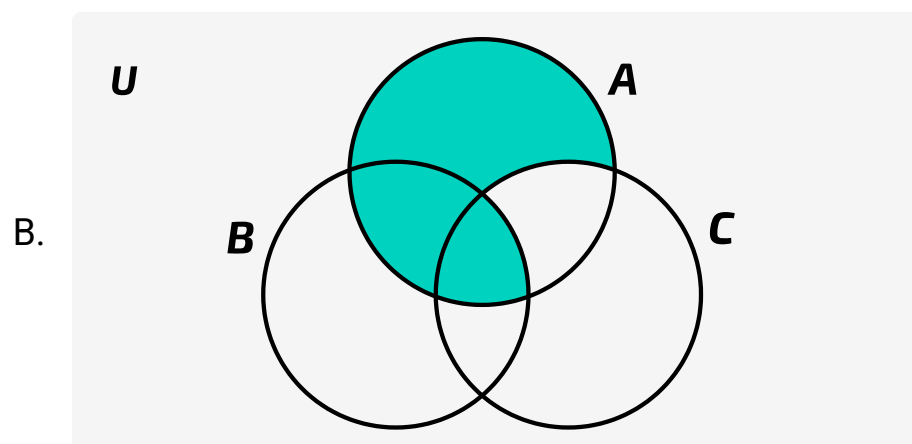
Set operations and venn diagrams 1

Advanced: Challenge 1

Which of these Venn diagrams denotes the operation $A \cap B \cap C$?

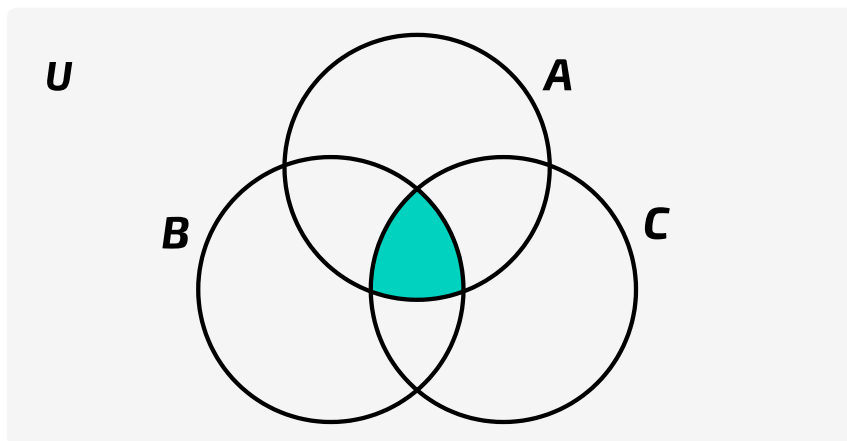


Option A



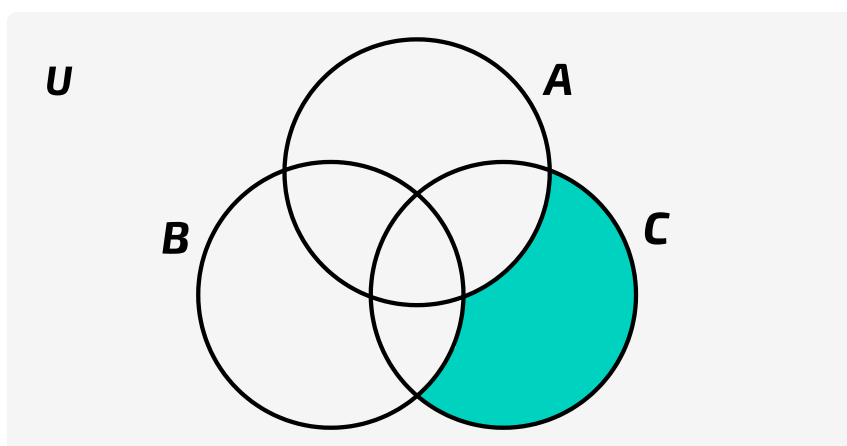
Option B

C.



Option C

D.



Option D

- ☐ A
- ☐ B
- ☐ C
- ☐ D

Quiz:

**STEM SMART 2026 Residential Extend Quiz 4 –
Sets and Functions**

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Sets: union 2

Advanced: Challenge 2  

Three sets have been defined:

$$A = \{2, 4, 6, 8, 10, 12\}$$

$$B = \{3, 6, 9, 12\}$$

$$C = \{4, 8, 11, 12\}$$

What is the cardinality (number of elements) of $A \cup B \cup C$?

Quiz:

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Sets: cartesian product 3

Advanced: Challenge 2  

Two sets have been defined:

$$A = \{1, 2, 3, 4\}$$

$$B = \{0, 3, 6\}$$

Which **three** of the tuples below are elements of the Cartesian product $A \times B$?

☐ $\{0, 6\}$

☐ $\{0, 1\}$

☐ $\{1, 3\}$

☐ $\{4, 3\}$

☐ $\{3, 4\}$

☐ $\{2, 6\}$

Quiz:

[STEM SMART 2026 Residential Extend Quiz 4 –](#)

[Sets and Functions](#)

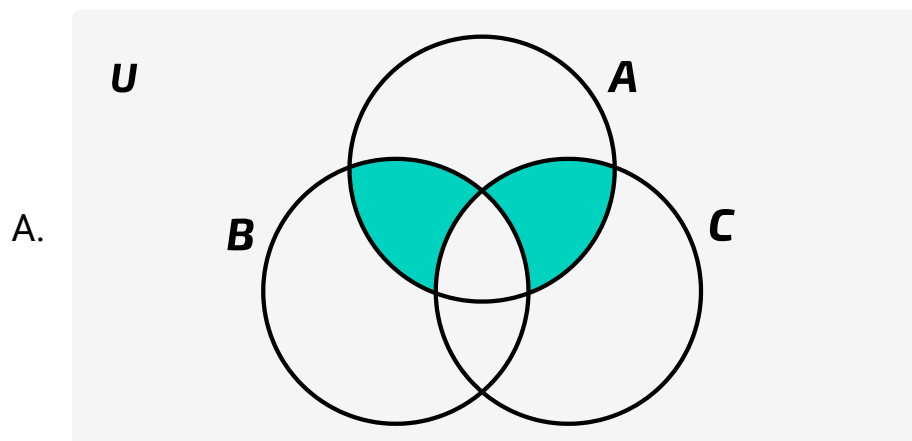
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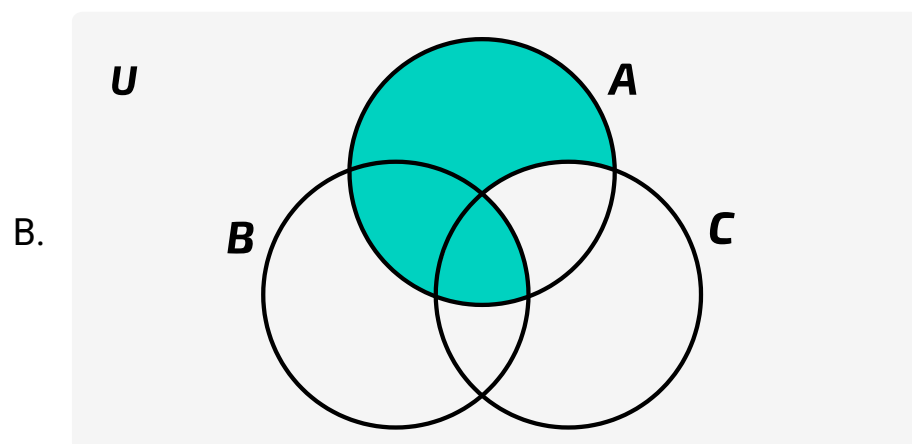
Set operations and venn diagrams 2

Advanced: Challenge 2 ■■

Which of these Venn diagrams denotes the operation $C \setminus (A \cup B)$?

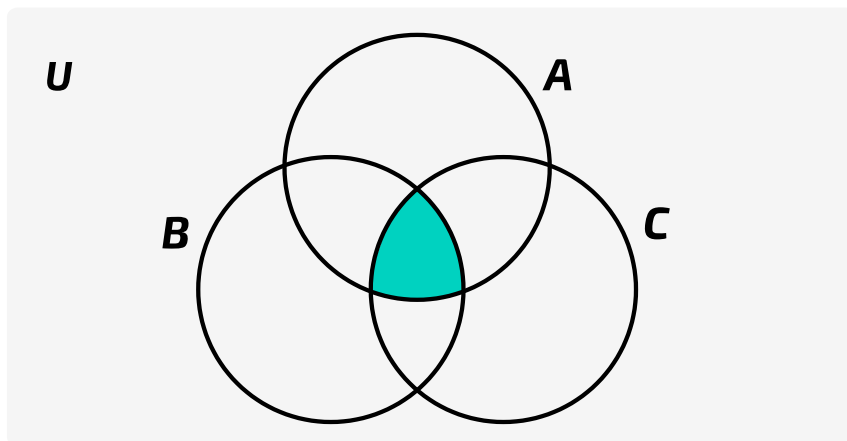


Option A



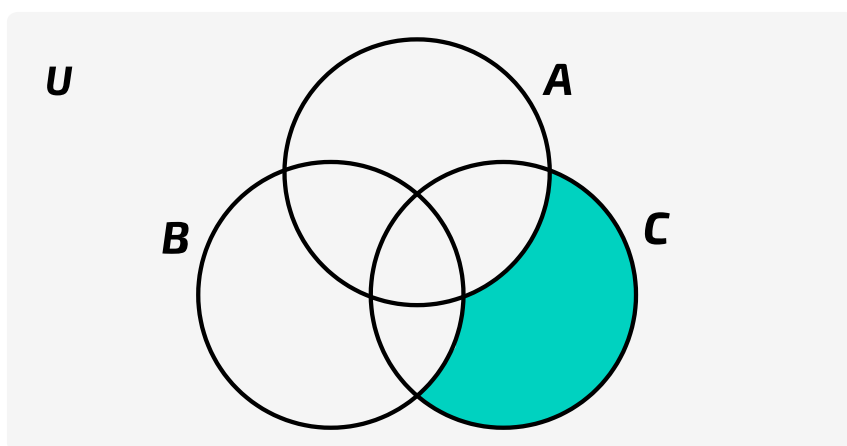
Option B

C.



Option C

D.



Option D

- ☐ A
- ☐ B
- ☐ C
- ☐ D

Quiz:

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Sets: which expression? 1

Advanced: Challenge 2  

Given three sets A , B , and C :

$$A = \{1, 3, 4, 6, 7\}$$

$$B = \{1, 5, 7\}$$

$$C = \{2, 4, 6, 7\}$$

Which of these expressions would return the empty set $\emptyset$?

☐ $A \cup B$

☐ $B \cup C$

☐ $B \setminus A$

☐ $(B \cap C) \setminus A$

Quiz:

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Sets: which operation? 2

Advanced: Challenge 2 

Given three sets A , B , and C :

$$A = \{2, 3, 4, 5\}$$

$$B = \{2, 3, 4, 5\}$$

$$C = \{3, 5\}$$

Which **three** of these statements are correct?

☐ $C \subset B$

☐ $A \subseteq B$

☐ $C \subset A$

☐ $B \subset A$

Quiz:

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Sets and Functions](#)

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Exponential functions

Advanced: Practice 2  

Which of the following is an exponential function?

☐ $f(x) = 3 \log_{10} x$

☐ $f(x) = 2x^3 + 1$

☐ $f(x) = 4x + 1$

☐ $f(x) = 3^x + 5$

Quiz:

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Sets and Functions**

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